

1. Computer Will Not Power On

Situation

You press the power button on a desktop computer, but nothing happens. There are no lights, no fan movement, and no display.

How a Multimeter Helps

A multimeter is used to check whether the **Power Supply Unit (PSU)** is providing the correct voltage to the computer.

Steps

- Set the multimeter to **DC Voltage** mode.
- Measure the voltage on the PSU connectors.
- Check common output voltages:
 - **+12V** (Yellow wire)
 - **+5V** (Red wire)
 - **+3.3V** (Orange wire)
 - **Ground** (Black wire)
- Compare the measured values with the standard voltage ratings.

Possible Results

- **Correct voltage:** The PSU is working, so the problem may be the motherboard, RAM, or CPU.
- **No voltage or incorrect voltage:** The PSU is faulty and should be repaired or replaced.

2. Smartphone Will Not Charge

Situation

A smartphone does not charge even when connected to a charger.

How a Multimeter Helps

A multimeter helps determine whether the problem is with the charger, USB cable, charging port, or battery.

Steps

- Test the charger output voltage.
 - A standard USB charger should provide approximately **5V DC**.
- Check the USB cable for continuity to ensure the wires are not broken.
- Measure the battery voltage (if the battery is removable or accessible).
 - A healthy lithium-ion battery typically measures around **3.7V–4.2V**.
- Inspect the charging port if the charger and battery are functioning correctly.

Possible Results

- **No charger voltage:** Replace the charger.
- **Broken cable:** Replace the USB cable.
- **Low battery voltage:** The battery may be deeply discharged or damaged.
- **All tests normal:** The charging IC or charging port may require repair.

3. Detecting a Short Circuit in a Laptop or Smartphone

Situation

A laptop or smartphone becomes hot quickly, shuts down unexpectedly, or does not turn on because of a possible short circuit.

How a Multimeter Helps

The multimeter is used in **continuity** or **resistance** mode to identify short circuits or damaged connections.

Steps

- Turn off the device and disconnect the battery or power source.
- Set the multimeter to **continuity** or **resistance (Ω)** mode.
- Place the probes across suspected components or power lines.
- Listen for the continuity beep or observe the resistance reading.

Possible Results

- **Continuity beep or very low resistance:** A short circuit may exist.

- **No continuity where there should be a connection:** A broken trace, wire, or fuse may be present.
- Replace or repair the faulty component after identifying the problem.